

# TRIAGE-LINK — Commercialization & Go-to-Market Strategy

**Purpose.** A practical plan to take TRIAGE-LINK from a working prototype to a **commercial product**, with a clear beachhead, business model, productization plan, and phased roadmap. It complements two existing docs: `docs/canada-market-survey.md` (the competitive/regulatory landscape) and `docs/CERTIFICATION-ROADMAP.md` (the medical-device pathway). This doc is the **business** layer those two imply.

⚠ **Not legal, financial, or regulatory advice.** Engage a regulatory-affairs professional, a privacy/health-law advisor, and a sales/commercial lead before committing. Figures are planning estimates, not quotes.

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## 1. Executive summary

TRIAGE-LINK is an **offline-first, no-backend PWA** that uniquely combines **prehospital ePCR + MCI/START triage + FHIR R4 handover**, multilingual (EN/FR/AR/FA incl. RTL). The Canadian market survey shows the **official provincial/municipal EMS ePCR market is a fortress** — mandatory Ontario OADS v4.0 compliance, entrenched cloud incumbents (ESO/Interdev-iMedic, ESO/Medusa-Siren, ImageTrend), multi-year RFP cycles — and even the displaced Prehos customers are migrating to *other cloud incumbents*, not offline-first tools.

**Recommendation: do not fight that fortress first.** Enter through an **underserved beachhead** where TRIAGE-LINK's exact strengths are decisive and the regulatory burden is lighter — **humanitarian / NGO / global-health and disaster / MCI / event medicine**. Frame the product as a **documentation & coordination tool** (not clinical-decision SaMD) to stay on the lighter regulatory path, land **2-3 reference deployments**, build evidence and revenue, and only take on the full **SaMD + OADS/NEMSIS** burden when expanding toward the formal EMS channel.

**The wedge in one line:** *the only field casualty tool that runs with zero connectivity and zero server, speaks four languages including RTL, and does triage + charting + standards-based handover in one installable app — at a fraction of the per-seat cost of cloud ePCR.*

## 2. What we have to sell (assets & differentiation)

**Product assets (built today):**

- Offline-first PWA — installable, works fully offline, deploys as static files (no backend required).
- Capability breadth in one app: anatomical **body-chart injury capture**, timestamped **vitals** (+GCS) with **trend sparklines**, **treatments**, **START triage** + multi-casualty **triage board**, **time-since-injury** clocks, **incident-command scene summary**, **AT-MIST casualty card**, **FHIR R4** export (Patient + Encounter + Provenance).
- **Multilingual EN/FR/AR/FA** with full **RTL** — rare in this category.
- **At-rest encryption vault** (AES-256-GCM) + encrypted backups; security-hardened (CSP, API hardening, CI scanning).
- Clean, framework-free domain core ( [@triage-link/core](#) ) reusable by future native/back-end clients.

### Differentiation vs the surveyed Canadian field:

| Capability                                          | TRIAGE-LINK | ESO/Interdev (iMedic) | ESO/Medusa (Siren) | ImageTrend     |
|-----------------------------------------------------|-------------|-----------------------|--------------------|----------------|
| ePCR                                                | ✓           | ✓                     | ✓ deep             | ✓ deep         |
| MCI/START triage + board                            | ✓           | not advertised        | not advertised     | not advertised |
| FHIR R4 handover                                    | ✓ named     | generic only          | not advertised     | ✓ (Hub)        |
| <b>No backend / no sync server</b>                  | ✓           | ✗ cloud               | 🔕 store-and-sync   | ✗ SaaS         |
| Multilingual + <b>RTL</b>                           | ✓           | not advertised        | not advertised     | not advertised |
| Zero per-seat infrastructure                        | ✓           | ✗                     | ✗                  | ✗              |
| <b>Provincial-standard (OADS/NEMSIS) compliance</b> | ✗ (gap)     | ✓                     | ✓                  | ✓              |
| CAD/dispatch + EHR integration                      | ✗ (gap)     | ✓                     | ✓                  | ✓              |
| Mature ePCR depth, billing, analytics               | ✗ (gap)     | ✓                     | ✓                  | ✓              |

*Read:* we win on **portability, breadth-in-one-app, language, and cost**; incumbents win on **depth, integration, and compliance**. The strategy is to sell into contexts where the first column matters more than the last three rows.

## 3. Strategic thesis

1. **The official-EMS lane is the hardest possible first market.** It requires OADS/NEMSIS compliance, CAD/EHR integration, an ISO 13485 QMS, and survives

only through multi-year procurement against incumbents with reference bases. That's a capital-heavy, slow fight to *match* incumbents on *their* strengths.

2. **Our strengths are someone else's unmet needs.** Offline/no-infrastructure, multilingual-RTL, and combined MCI+ePCR are exactly what humanitarian, disaster, austere, and event-medicine contexts need — and what cloud ePCRs are *bad* at.
3. **Lighter regulatory path is available** if intended use is framed as **documentation/coordination**, not clinical decision-making (see [CERTIFICATION-ROADMAP.md](#) §1). That collapses time-to-first-revenue from years to months.
4. **Land-and-expand:** prove value + safety + references in the niche, then climb toward regulated/EMS markets with a product that already has users, evidence, and revenue behind it.

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## 4. Target segments & beachhead

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Scored on fit (does our differentiation win?), accessibility (can a small team reach buyers?), regulatory burden, and willingness to pay.

| Segment                                   | Differentiation fit | Accessibility | Reg. burden | Verdict              |
|-------------------------------------------|---------------------|---------------|-------------|----------------------|
| Humanitarian / NGO / global health        | ★★★                 | ★★            | ★ (light)   | Primary beachhead    |
| Disaster / MCI & event medicine           | ★★★                 | ★★★           | ★★          | Primary beachhead    |
| Remote-industrial / expedition / maritime | ★★                  | ★★            | ★★          | Secondary            |
| Training & simulation (paramedic schools) | ★★                  | ★★★           | ★ (light)   | Early-traction wedge |
| Official EMS (provincial/municipal)       | ★                   | ★             | ★★★ (heavy) | Phase 3 expansion    |

**Beachhead = Humanitarian/NGO/global-health + Disaster/MCI/event medicine.**

They share the same buyer pains and the same product fit, so one product configuration serves both.

- **Why they fit:** intermittent or absent connectivity; multilingual incl. Arabic/Persian populations; multi-casualty scenes; no IT infrastructure to deploy a server; tight budgets; need a printable/shareable handover artifact.
  - **Who buys:** international NGOs and their field medical programs, national Red Cross/Red Crescent societies, disaster-response and emergency-management agencies, event-medical providers (festivals/sports/marathons), and global-health implementers. (Defence/TCCC is adjacent but out of this plan's scope.)
  - **Training/simulation** is a fast, low-risk way to seed reference users and clinical champions while the field deployments mature.
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## 5. Positioning & value proposition

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**Positioning statement:** *For field medical teams working where connectivity, infrastructure, and language can't be assumed, TRIAGE-LINK is the install-anywhere casualty-documentation and triage app that captures, triages, and hands over patient care in one offline tool — unlike cloud ePCR platforms that need servers, single-language UIs, and per-seat contracts.*

**Value pillars:**

1. **Works when nothing else does** — zero connectivity, zero server, installs on any device.
2. **One tool, whole workflow** — triage board + charting + FHIR handover + scene summary.
3. **Speaks your patients' language** — EN/FR/AR/FA with RTL, extensible.
4. **Owns its data** — on-device, encrypted, exportable; no vendor lock-in or cloud-continuity risk (cf. the Prehos collapse).
5. **Radically lower cost** — no per-seat cloud bill; deploy to a whole team for a fraction of a ~\$60K/yr ePCR contract.

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## 6. Business model & pricing

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**Licensing decision (do this first):** the repo is **MIT** today. For a commercial product, move to an **open-core / dual-license** posture: keep a free community core (drives humanitarian goodwill, adoption, and trust), and gate **commercial value-adds** (multi-tenant admin, SSO/RBAC, audit/compliance pack, support SLA, integrations, hosted sync) behind paid tiers. *Decide before external contributions and customer deployments accrue.*

**Tiering (illustrative):**

| Tier                    | Audience             | What's included                                                               | Model                            |
|-------------------------|----------------------|-------------------------------------------------------------------------------|----------------------------------|
| <b>Community</b>        | Solo/NGO/training    | Full offline PWA, self-hosted static deploy                                   | Free / open-source               |
| <b>Team</b>             | Event/NGO programs   | Multi-device admin, training, priority support, branded builds                | Annual subscription per org/site |
| <b>Organization</b>     | Larger NGOs/agencies | SSO/RBAC, audit log, optional hosted sync, data-residency options, onboarding | Annual; volume tiers             |
| <b>Enterprise / EMS</b> | Formal EMS (Phase 3) | OADS/NEMSIS conformance, CAD/EHR integration, certified SaMD, QMS docs, SLA   | Contract / RFP                   |

**Pricing strategy:** lead with **cost disruption**. The no-backend architecture removes the per-seat cloud cost that anchors incumbents (~\$63K/yr at a mid-size service). Price

the **Team/Org** tiers as a **site/program licence + support**, not per-seat, so a 50-person event team or NGO field program is dramatically cheaper than cloud ePCR. For humanitarian buyers, expect **grant-funded** or **free-core + paid-support/services** revenue rather than SaaS seats.

**Revenue logic:** services and support (deployment, training, custom languages, integration, compliance evidence) will likely out-earn licences in the niche phase — and they build the references and case studies that unlock the bigger markets later.

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## 7. Regulatory & compliance posture (beachhead)

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- **Intended use = documentation & coordination tool**, explicitly *not* a clinical-decision SaMD (see `CERTIFICATION-ROADMAP.md` §1). Write the **Intended Use / Indications** statement to match, and make sure computed outputs (TBSA, START category) are presented as **user-entered/clinician-confirmed records**, not guidance. This keeps the beachhead on the **light regulatory path**.
  - **Privacy & data protection (must-have to handle real PHI):** even as a documentation tool, you handle PHI. Establish: a privacy policy + DPA, **data-residency** options, breach-notification process, and alignment with **PHIPA/PIPEDA** (Canada) and **GDPR** (if EU/EEA data) — and note humanitarian deployments add their own data-protection norms (e.g., ICRC data-protection standards).
  - **Security productization:** flip the encryption vault from opt-in to **always-on by policy** for org tiers; add **RBAC, SSO, immutable audit log**, retention/destruction. Pursue **SOC 2 Type II** + an independent **pen test** when org/enterprise deals require it.
  - **Defer the heavy stack** (ISO 13485 QMS, IEC 62304/62366/14971 full evidence, OADS/NEMSIS conformance, Health Canada/FDA submission) to **Phase 3**, when chasing regulated/EMS revenue — per the cert roadmap's phased plan.
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## 8. Productization gaps (prototype → commercial-ready)

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**Must-have for the beachhead (Phase 1):**

- **Accounts & multi-tenant org management** (today it's single-device).
- **Always-on encryption policy, RBAC, SSO** (org tier), **immutable audit log**.
- **Optional hosted sync** (productionize `packages/sync-service`: auth, multi-tenant, DR/backup, observability) — *optional*, because offline-only must remain first-class.
- **Admin & reporting** (export, roster, basic analytics), **data import/export/migration**.
- **Accessibility (WCAG), device/MDM** guidance, **app-store/installable** distribution polish.
- **Onboarding, training material, and support** processes + documentation.
- **Language extensibility** (add a language without a code release) — a direct selling point.

**Later (Phase 2-3):** OADS/NEMSIS dataset conformance, CAD/dispatch + hospital-EHR integration, billing, deep analytics, certified-SaMD evidence.

Engineering note: keep **offline-first and no-backend as the default**. Every paid feature (sync, SSO, audit) must degrade gracefully to the offline core — that's the moat, not a checkbox.

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## 9. Go-to-market motion

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1. **Design-partner pilots (2-3).** Recruit a national Red Cross/Red Crescent program, a disaster-response/emergency-management agency, and an event-medical provider. Offer free/subsidized deployment + hands-on support in exchange for feedback, a reference, and a case study.
  2. **Land via champions, not RFPs.** In the niche, adoption is clinician/coordinator-led, not procurement-led. Win a field medical lead or event medical director; let usage spread.
  3. **Evidence flywheel.** Each pilot → usage data + a published case study + (ideally) a small field/simulation study (cf. the KatApp result that app-triage beats paper). Evidence de-risks the next sale and seeds the eventual SaMD clinical file.
  4. **Partnerships:** humanitarian tech consortia, FHIR/interop communities, training organizations, and device makers — channels that reach the buyers a small team can't cold-call.
  5. **Content & community:** open-source core + visible roadmap builds trust in a sector that distrusts vendor lock-in (Prehos is the cautionary tale to reference).
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## 10. Phased commercial roadmap

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| Phase                                   | Goal                          | Key work                                                                                                                 | Exit criteria                                                  |
|-----------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>0 — Validate (0-3 mo)</b>            | Confirm demand & intended-use | Intended-use statement; 8-12 buyer interviews; licensing decision (open-core); privacy posture; pick 2-3 design partners | Signed design partners; go/no-go                               |
| <b>1 — Beachhead pilots (3-9 mo)</b>    | First real deployments        | Phase-1 productization (accounts, RBAC, audit, always-on crypto, support); deploy + support pilots; collect evidence     | 2-3 live deployments; 1 case study; first paid support/licence |
| <b>2 — Scale the niche (9-24 mo)</b>    | Repeatable revenue in niche   | Org tier + optional hosted sync; partnerships; pricing/packaging; SOC 2 if needed; small field/sim study                 | Repeatable sales motion; reference base; sustainable revenue   |
| <b>3 — Regulated expansion (24 mo+)</b> | Enter EMS / regulated markets | SaMD path (ISO 13485, IEC 62304/62366/14971), OADS/NEMESIS conformance, CAD/EHR integration, Health Canada/FDA           | Certified product; first official EMS contract                 |

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## 11. Team, org & funding

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- **Founding gaps to fill: regulatory/quality** lead (fractional at first), a **clinical lead/advisor** (paramedic/EM physician) for credibility + the advisory board, and **sales/BD** for the niche. Engineering exists; these don't.
  - **Company & risk:** incorporate; obtain **product/professional liability insurance** appropriate to medical software; vulnerability-disclosure policy.
  - **Funding fit:** the niche-first path suits **non-dilutive funding** — health-innovation grants, humanitarian-tech funds, global-health/disaster-resilience programs, and government innovation programs — better than VC. Reserve VC/large raises for Phase 3 (regulated expansion), where capital intensity justifies it.
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## 12. Risks & mitigations

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| Risk                                                                                                               | Mitigation                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Intended-use creep into SaMD</b><br>(TBSA/START presented as guidance)<br>triggers heavy regulation prematurely | Lock the documentation-tool intended-use; present computed values as confirmable records; legal review                    |
| <b>PHI/privacy exposure</b> in a security-sensitive sector                                                         | Always-on encryption, audit log, data residency, DPA, breach process before any real-PHI pilot                            |
| <b>No clinical evidence / references yet</b>                                                                       | Design-partner pilots + a small field/simulation study early; publish                                                     |
| <b>Open-source core cannibalizes revenue</b>                                                                       | Open-core boundary drawn at org/compliance/support features; services-led revenue in niche                                |
| <b>Niche is small / low willingness-to-pay</b>                                                                     | Services + grants + multi-segment (NGO+disaster+events+training) to aggregate demand; niche is the wedge, not the ceiling |
| <b>Incumbents add offline/multilingual</b>                                                                         | Move fast on references; the combined no-backend + RTL + MCI set is hard to retrofit onto cloud SaaS                      |
| <b>Solo-vendor trust concern</b> (cf. Prehos)                                                                      | Open-source core + data-ownership story turns the sector's fear into our advantage                                        |

## 13. What success looks like (KPIs by phase)

- **Phase 0:** ≥3 signed design partners; documented intended-use + licensing decision.
- **Phase 1:** 2-3 live field deployments; 1 published case study; first revenue (support/licence); zero PHI incidents.
- **Phase 2:** repeatable sales motion; named reference customers across ≥2 sub-segments; 1 field/simulation study; positive gross margin on services + licences.
- **Phase 3:** certified product; first official EMS/regulated contract.

## 14. Immediate next steps (30 / 60 / 90 days)

- **30 days:** draft the **Intended Use / Indications** statement; make the **open-core licensing** decision; write a one-page **privacy & data-protection posture**; build a target list of 15-20 design-partner candidates.
- **60 days:** 8-12 **buyer-discovery interviews**; sign **2-3 design partners**; scope **Phase-1 productization** backlog (accounts, RBAC, audit, always-on crypto, support).
- **90 days:** begin Phase-1 build; stand up support/onboarding; line up a **clinical advisor**; identify 1-2 **non-dilutive funding** programs to apply to.



## Appendix — sources & cross-references

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- Competitive & regulatory landscape: [docs/canada-market-survey.md](#) (incl. the OADS v4.0 compliance moat, ESO/Interdev/Medusa/ImageTrend footprints, and the Prehos collapse).
- Medical-device pathway & standards stack: [docs/CERTIFICATION-ROADMAP.md](#) (intended-use decision, ISO 13485 / IEC 62304 / 62366 / 14971 / 81001-5-1, phased certification, indicative cost & timeline).
- Engineering build plan: [docs/ROADMAP.md](#).

*This is a working strategy, not a commitment or forecast. Validate every assumption with real buyer discovery before investing.*